THE IDENTITY COMPONENT G^0 OF A GROUP SCHEME OVER A FIELD IS A FLAT CLOSED GEOMETRICALLY IRREDUCIBLE QUASI-COMPACT SUBGROUP SCHEME

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Allowed standard prerequisites (editorial).

- Definition morphisms-definition-scheme-structure-connected-component (canonical tag 04PX). cited at proof lines 1243-1246. The canonical scheme structure on the connected component.
- Lemma lemma-closed-subgroup-scheme (canonical tag 0G8L). cited at proof lines 1246-1249. Criterion for closed subgroup scheme.
- Varieties lemmas 056R and 0385. cited at proof lines 1252-1256. Geometric connectedness and connectedness of the square.
- Lemma morphisms-lemma-characterize-flat-closed-immersions (canonical tag 04PW). cited at proof lines 1258-1260. Factorization of m through G^0.
- Lemmas 0B7Q, 047M, 0B7P. cited at proof lines 1261-1266. Irreducibility, geometric irreducibility, quasi-compactness.

Source reference. Original definition 04PX: sources/stacks-chow/tags.txt, source lines 6130–6130. Cited at proof lines 1243-1246 to define G^0; navigation for the setup.

Source reference. Original lemma 0G8L: sources/stacks-chow/tags.txt, source lines 20271–20271. Cited at proof lines 1246-1249 as the criterion used to prove G^0 is a closed subgroup scheme; load-bearing.

Source reference. Original lemma 056R: sources/stacks-chow/tags.txt, source lines 6719–6719. Cited at proof lines 1252-1254 to see G^0 is geometrically connected; load-bearing.

Source reference. Original lemma 0385: sources/stacks-chow/tags.txt, source lines 4224–4224. Cited at proof lines 1254-1256 to see G^0 x_k G^0 is connected; load-bearing.

Source reference. Original lemma 04PW: sources/stacks-chow/tags.txt, source lines 6129–6129. Cited at proof lines 1258-1260 to factor m through G^0; load-bearing.

Source reference. Original lemma 0B7Q: sources/stacks-chow/tags.txt, source lines 14113–14113. Cited at proof lines 1261-1262 for irreducibility of G^0; load-bearing.

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Source reference. Original lemma 047M: sources/stacks-chow/tags.txt, source lines 5513–5513. Cited at proof lines 1262-1264 for geometric irreducibility; load-bearing.

Source reference. Original lemma 0B7P: sources/stacks-chow/tags.txt, source lines 14112–14112. Cited at proof lines 1264-1266 for quasi-compactness; load-bearing.

Proposition 0.2. *Let G be a group scheme over a field k. There exists a canonical closed subgroup scheme $G^0 \subset G$ with the following properties*

(1) *$G^0 \to G$ is a flat closed immersion,*
(2) *$G^0 \subset G$ is the connected component of the identity,*
(3) *G^0 is geometrically irreducible, and*
(4) *G^0 is quasi-compact.*

Proof. Let G^0 be the connected component of the identity with its canonical scheme structure (Morphisms, Definition 04PX). To show that G^0 is a closed subsgroup scheme we will use the criterion of Lemma 0G8L. The morphism $e : \mathrm{Spec}(k) \to G$ factors through G^0 as we chose G^0 to be the connected component of G containing e. Since $i : G \to G$ is an automorphism fixing e, we see that i sends G^0 into itself. By Varieties, Lemma 056R the scheme G^0 is geometrically connected over k. Thus $G^0 \times_k G^0$ is connected (Varieties, Lemma 0385). Thus $m(G^0 \times_k G^0) \subset G^0$ set theoretically. Thus $m|_{G^0 \times_k G^0} : G^0 \times_k G^0 \to G$ factors through G^0 by Morphisms, Lemma 04PW. Hence G^0 is a closed subgroup scheme of G. By Lemma 0B7Q we see that G^0 is irreducible. By Lemma 047M we see that G^0 is geometrically irreducible. By Lemma 0B7P we see that G^0 is quasi-compact. □